UNIVERSITATEA BABEŞ-BOLYAI CLUJ-NAPOCA
FACULTATEA DE MATEMATICĂ ŞI INFORMATICĂ
SPECIALIZAREA Informatică

LUCRARE DE LICENŢĂ

Design and Implementation of an AI-Based Mobile Application for Automatic Gym Exercise Posture Analysis and Feedback

Conducător științific
Dr. Molnar Árthur

Absolvent
Strugari Cristian-Silviu

2026

ABSTRACT

This thesis presents the design and implementation of PoseFix, an AI-based mobile application for automatic gym exercise posture analysis and feedback. Chapter 1 introduces the problem of incorrect exercise execution among amateur athletes, motivates the need for an automated solution, and outlines the structure of the work. Chapter 2 provides the theoretical background, covering pose estimation techniques, the MediaPipe BlazePose model, and a review of related work. Chapter 3 describes the system architecture, including the three-tier design comprising a Flutter mobile application, a Java Spring Boot backend, and a Python machine learning service, as well as the user journey and database design. Chapter 4 presents the implementation details of each system component. Chapter 5 covers testing and performance evaluation. Chapter 6 draws conclusions and outlines directions for future work. The primary contribution of this thesis is a working end-to-end system capable of analysing a recorded gym exercise video and returning a structured posture score, detected mistakes, and joint angle measurements.

Cuprins

1 Introduction **1**

1.1 Problem Context 1

1.2 Proposed Solution 1

1.3 Technical Approach 2

1.4 Structure of the Thesis 2

1.5 Declaration of Generative AI and AI-Assisted Technologies in the Writing Process 2

2 Theoretical Background **3**

2.1 Pose Estimation in Computer Vision 3

2.1.1 2D vs. 3D Pose Estimation 3

2.1.2 Detection Approaches 4

2.2 MediaPipe BlazePose 4

2.2.1 Architecture 5

2.2.2 Coordinate System and Landmark Indices 5

2.3 Joint Angle Computation 5

2.4 Related Work 6

3 System Architecture and Design **7**

3.1 High-Level Architecture 7

3.1.1 Technology Stack 8

3.2 User Journey 8

3.3 Database Design 9

4 Titlul capitolului **11**

5 Concluzii **13**

Bibliografie **14**

Capitolul 1

Introduction

1.1 Problem Context

The incorrect execution of a gym exercise is one of the main causes of injuries in amateur athletes. Most individuals train without professional supervision, leading to improper posture, insufficient range of motion, or joint misalignment. While personal trainers provide real-time correction, their services are not always accessible or affordable. Therefore, there is a growing need for intelligent systems capable of automatically analysing exercise postures and providing corrective feedback [S+25] [S+24].

Common errors observed in resistance training include excessive forward lean of the torso, asymmetrical movement of the limbs, and insufficient range of motion through the targeted joints. These deviations are frequently invisible to the athlete without external observation and tend to accumulate into chronic injury over time. Studies indicate that the shoulder, lower back, and knee are the most frequently affected anatomical regions in gym-related injuries, with improper technique being a primary contributing factor [S+25].

The demand for scalable, cost-effective alternatives to personal training has grown alongside the popularisation of mobile technology and computer vision. Pose estimation models now operate in real time on consumer hardware, opening the possibility of providing instant biomechanical feedback through a smartphone application.

1.2 Proposed Solution

This thesis proposes the design and implementation of an AI-based mobile application capable of analysing gym exercises using pose estimation techniques. The system allows users to upload a recording of themselves performing an exercise,

after which computer vision algorithms extract the body key points and compute joint angles. Based on predefined rules and posture evaluation logic, the application detects common mistakes and provides structured feedback for improvement.

The proposed system, named **PoseFix**, aims to simulate the core corrective function of a personal trainer in an accessible and affordable digital format. By processing a video recording rather than requiring live streaming, the system accommodates real-world network constraints and allows users to record exercises at their own pace and in any gym environment.

1.3 Technical Approach

The application is developed using a Flutter-based mobile interface, a Java backend for business logic and data management, and a separate Python-based machine learning service responsible for posture analysis. Pose estimation models such as MediaPipe BlazePose [Tet16] or similar lightweight neural networks are integrated into the ML service to detect body landmarks from video frames. The system stores only relevant metadata in a Supabase database, while video data is processed temporarily and discarded after the analysis. The implementation includes requirement specification, system architecture design, algorithmic posture evaluation, testing, and performance validation.

1.4 Structure of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 provides the theoretical background on pose estimation, the MediaPipe BlazePose model, and related work in automated exercise coaching. Chapter 3 describes the overall system architecture, the user journey, and the database design. Chapter 4 presents the implementation of each system component in detail. Chapter 5 covers testing methodology and performance results. Chapter 6 draws conclusions and identifies future development directions.

1.5 Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the author used Claude (Anthropic) in order to assist with phrasing, structural suggestions, and code review during the development phase. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the thesis.

Capitolul 2

Theoretical Background

This chapter presents the theoretical foundations underlying the PoseFix system. It begins with an overview of pose estimation as a computer vision discipline, followed by a detailed description of the MediaPipe BlazePose model used in this work. The chapter concludes with a review of related work in the field of automated exercise analysis.

2.1 Pose Estimation in Computer Vision

Pose estimation is the computer vision task of detecting and localising the key anatomical landmarks of the human body — typically joints such as shoulders, elbows, wrists, hips, knees, and ankles — from image or video input. The output is commonly represented as a set of coordinates, called a skeleton or pose graph, describing the spatial position of each landmark.

2.1.1 2D vs. 3D Pose Estimation

Early approaches to pose estimation operated exclusively in two dimensions, producing (x, y) coordinates for each landmark within the image plane. Two-dimensional methods are computationally inexpensive and sufficient for many tasks, but they suffer from a fundamental limitation: depth ambiguity. Because the image is a projection of three-dimensional space onto a flat sensor, two different three-dimensional configurations can produce identical two-dimensional observations. This is particularly problematic in exercise analysis, where subtle out-of-plane deviations — such as a slight lean of the torso — may be invisible in the image plane.

Modern models, including MediaPipe BlazePose, estimate a third coordinate z, approximating the depth of each landmark relative to the centre of the body. While this estimate is less accurate than dedicated depth-camera measurements, it provides sufficient information to detect gross postural errors such as excessive back lean

and arm asymmetry. In this work, three-dimensional angle computation is used for all critical joint evaluations, as described in Chapter 4.

2.1.2 Detection Approaches

Pose estimation methods broadly fall into two categories: top-down and bottom-up. Top-down approaches first detect a bounding box around each person in the image and then apply a separate pose estimation model within each box. This produces high accuracy when individuals are well separated but is sensitive to missed detections. Bottom-up approaches detect all landmarks in the image simultaneously and then associate them into individual skeletons. Bottom-up methods scale better to crowded scenes but may produce lower per-person accuracy.

MediaPipe BlazePose uses a top-down pipeline optimised for single-person video, which is the relevant scenario for self-recorded gym exercise videos. A lightweight detector locates the person in the first frame, and a tracking model maintains the skeleton estimate across subsequent frames without re-running full detection on every frame. This design enables real-time performance on mobile hardware.

2.2 MediaPipe BlazePose

MediaPipe BlazePose is an open-source pose estimation model developed by Google and released as part of the MediaPipe framework. It produces 33 three-dimensional landmarks covering the full body, including the face, torso, arms, and legs. Each landmark is annotated with a visibility score indicating the model's confidence that the point is visible and unoccluded in the frame.

Figura 2.1: The 33 body landmarks produced by MediaPipe BlazePose, covering the full body skeleton.

2.2.1 Architecture

BlazePose consists of two neural network stages. The first stage is a lightweight person detector based on a single-shot detection architecture, responsible for localising the subject and providing a region of interest. The second stage is a regression network that maps the cropped image region to the 33 landmark coordinates and their visibility scores. The regression approach — predicting coordinates directly rather than generating heatmaps — enables sub-pixel accuracy and low latency.

In video mode, which is the mode used in this project, the framework skips the person detector on all frames after the first by using the previous frame's landmark positions to estimate the region of interest for the current frame. This tracking-based approach significantly reduces computational load and enables smooth landmark trajectories, which is important for reliable angle calculations over time.

2.2.2 Coordinate System and Landmark Indices

Each landmark is represented as a normalised coordinate (x, y, z) where x and y are normalised to the image width and height respectively, and z represents depth relative to the midpoint of the hips. The sign convention for z places landmarks closer to the camera at more negative values. The 33 landmarks are indexed from 0 to 32, with the following indices relevant to the exercises analysed in this project:

Index	**Side**	**Landmark**
11	Left	Shoulder
12	Right	Shoulder
13	Left	Elbow
14	Right	Elbow
15	Left	Wrist
16	Right	Wrist
23	Left	Hip
24	Right	Hip
25	Left	Knee
26	Right	Knee

Tabela 2.1: BlazePose landmark indices used in the posture evaluation logic of PoseFix.

2.3 Joint Angle Computation

A joint angle is defined as the angle at vertex point B formed by the vectors $\overrightarrow{BA}$ and $\overrightarrow{BC}$, where A, B, and C are three landmark coordinates. In three-dimensional space, this is computed using the dot product formula:

$$\theta = \arccos\left(\frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{\left\|\overrightarrow{BA}\right\| \cdot \left\|\overrightarrow{BC}\right\|}\right) \tag{2.1}$$

The result θ is expressed in degrees and represents the opening angle at the joint. For example, the back angle is computed with the left shoulder as point A, the left hip as vertex B, and the left knee as point C. A value near 180° indicates an upright posture, while a value significantly below 90° indicates excessive forward or backward lean.

Numerical stability is ensured by clipping the cosine value to the range $[-1, 1]$ before applying $\arccos$, preventing domain errors due to floating-point rounding.

2.4 Related Work

The use of computer vision for exercise analysis has attracted growing research interest in parallel with the improvement of real-time pose estimation models.

Sim et al. [S+24] proposed a virtual gym assistant using MediaPipe for pose estimation, demonstrating the feasibility of rule-based posture evaluation using landmark coordinates. Their system detected incorrect squat and bicep curl form using angle thresholds applied to 2D landmarks. The present work extends this approach to 3D coordinates and introduces a proportional scoring model rather than binary pass/fail judgements.

Tetreault [Tet16] studied behavioural interventions for reducing unsafe weightlifting techniques in a gym environment. While that work focused on behavioural coaching rather than computer vision, it established a quantitative taxonomy of common form errors in resistance training that informs the rule design in this project.

Shim et al. [S+25] provide a contemporary review of injury mechanisms in resistance training, identifying the shoulder, lumbar spine, and knee as the most commonly injured regions and attributing a significant proportion of injuries to technique errors. This motivates the specific checks implemented in the posture evaluation module, particularly the back angle and range-of-motion rules.

Compared to existing systems, PoseFix differs in three key respects: it operates on user-recorded video rather than live streaming, it uses a three-tier microservice architecture that cleanly separates UI, business logic, and ML concerns, and it returns a structured machine-readable result (score, mistake list, angle summary) rather than only a visual overlay.

Capitolul 3

System Architecture and Design

This chapter presents the overall design of the PoseFix system. It describes the high-level three-tier architecture, the user journey from registration through posture feedback, and the relational database schema.

3.1 High-Level Architecture

PoseFix is structured as a three-tier microservice system, with a clear separation of concerns between the mobile client, the backend server, and the machine learning service. Figure 3.1 illustrates the relationships between these components.

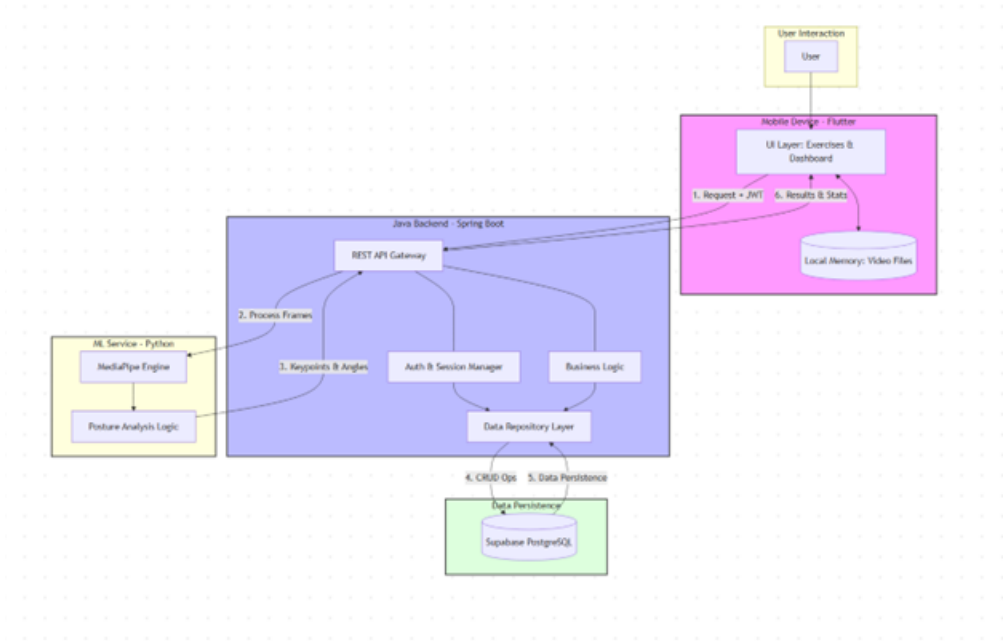

Figura 3.1: High-level architecture of the PoseFix system, showing the three-tier structure and the data flow between components.

The **Flutter mobile application** serves as the user interface layer. It communica-

tes exclusively with the Java backend over HTTP and relies on the Supabase client SDK only for authentication token management. The mobile client never contacts the Python ML service directly.

The **Java Spring Boot backend** is responsible for authentication enforcement, business logic, data persistence, and orchestration. When a user submits a video for analysis, the backend stores the file locally, forwards it to the Python service by reference, receives the result, persists it to the database, and returns it to the mobile client. The backend is also the sole layer that communicates with the Supabase PostgreSQL database.

The **Python FastAPI ML service** is a stateless microservice with a single responsibility: accepting a file path and an exercise identifier, processing the video using MediaPipe BlazePose, evaluating the pose against rule-based heuristics, and returning a structured JSON result. It holds no database connection and no user state.

Supabase provides two infrastructure services: the authentication layer, where user identity is managed and JWT tokens are issued, and the PostgreSQL database, where all application data is persisted.

This architecture offers several advantages relevant to the thesis context. The ML service can be developed, tested, and restarted independently of the backend. The backend can be extended with new business features without modifying the ML logic. The mobile client requires only knowledge of the backend API contract.

3.1.1 Technology Stack

Layer	Technology
Mobile	Flutter (Dart), Supabase Auth SDK, GoRouter, Provider
Backend	Java 17, Spring Boot 3.x, Spring Security, Spring Data JPA
ML Service	Python 3.11, FastAPI, MediaPipe Tasks API, OpenCV, NumPy
Database	Supabase (PostgreSQL 15), Row Level Security
Authentication	Supabase Auth (JWT)

Tabela 3.1: Technology stack used in the PoseFix system.

3.2 User Journey

Figure 3.2 presents the complete user journey through the PoseFix application, from initial registration to viewing posture feedback and tracking progress.

Figura 3.2: User journey through the PoseFix application, covering onboarding, workout execution, data processing, and progress tracking.

The journey is divided into four phases.

Onboarding covers initial registration via Supabase Auth, followed by the user entering personal data: name, birth date, height, weight, and sex. This data is stored in the `profiles` table and used for contextualisation of feedback in future versions.

Workout Execution is the core operational loop. The user selects an exercise from the library, logs the session parameters (sets, reps, load), records a video of themselves performing the exercise, and uploads the video for analysis. The system returns a posture score, a list of detected mistakes with severity ratings, and a summary of measured joint angles.

Data Processing occurs server-side and is transparent to the user. The backend receives the video, stores it temporarily, invokes the ML service, persists the result, and deletes the video file. Only the structured analysis metadata is retained in the database.

Progress and Tracking allows the user to review their workout history, track weight trends over time, and compare posture scores across sessions to observe improvement.

3.3 Database Design

The database schema consists of five tables managed by Supabase PostgreSQL. Figure 3.3 illustrates the entity-relationship structure.

profiles holds one record per user, linked to the Supabase `auth.users` table via the `user_id` primary key. It stores name, birth date, height, weight, and sex. An update trigger automatically maintains the `updated_at` timestamp.

weight_history records one weight measurement per user per calendar day, enforced by a unique index on `(user_id, measured_on)`. This design prevents multiple daily entries that could distort trend visualisations.

exercises is a static reference table defining the supported exercises by name, code, category, and muscle group. The `code` column (e.g. `lat_pulldown`) is used as a stable identifier for routing within the ML service.

workouts records individual training sessions. Each row is owned by a user and references an exercise. It stores the performed timestamp, duration, sets, reps, load, and optional notes.

posture_analyses holds the result of a video analysis. It is linked one-to-one with

Figura 3.3: Entity-relationship diagram of the PoseFix database schema.

a workout via a unique constraint on `workout_id`. The `mistakes` and `angles_summary` columns are stored as JSONB, allowing the flexible structure returned by the ML service to be persisted without a fixed schema.

The relationships between tables follow a clear ownership hierarchy:

$$\texttt{auth.users}1 : 1\texttt{profiles}1 : N\texttt{workouts}1 : 1\texttt{posture_analyses} \tag{3.1}$$

Weight history is independently owned by the user and does not participate in the workout chain.

Capitolul 4

Titlul capitolului

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla

pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum

Capitolul 5

Concluzii

Concluzii ...

Bibliografie

[S+24] J. H. Sim et al. Posture estimation and virtual gym assistant using MediaPipe. *International Research Journal of Modernization in Engineering Technology and Science (IRJMETS)*, 2024.

[S+25] Y. Shim et al. Most common injuries in resistance training: Mechanisms, risk factors, and prevention strategies. 2025.

[Tet16] C. L. Tetreault. Reducing performance of unsafe weight lifting techniques. Master's thesis, Western Michigan University, 2016.